

Part D

Answer any two questions from the following.

Each question carries 10 marks.

31. Explain Demand function.

32. (a) Cost function is given by $\pi = a + bq + cq^2$. Prove that $\frac{d(AC)}{dq} = \frac{MC - AC}{q}$.

(b) Given $TR = 1400q - 6q^2$ and $TC = 1500 + 80q$. Calculate the maximum profit.

33. Explain briefly the concept of marginal rate of substitution.

34. Find the critical values for minimizing the costs of a firm producing two goods x and y when the total cost function is $c = 8x^2 - xy + 12y^2$ and the firm is bound by contract to produce a minimum combination of goods totaling 42, that is, subject to the constraint $x + y = 42$.

(2 × 10 = 20 marks)

**FIRST SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 1C 01—MATHEMATICS

(2016—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type Questions)

Answer all questions (1-12).

Each question carries 1 mark.

1. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\frac{\pi}{2} - x} = \dots\dots\dots$
2. State sandwich theorem for limits.
3. What is a jump discontinuity ?
4. State Max-Min theorem for continuous functions.
5. Define point of inflection of a function $y = f(x)$.
6. What are the asymptotes of $y = \tan x$.
7. If $y = x^4 - 3 \cos x + e^x$, $dy = \dots\dots\dots$
8. Find the critical points of $f(x) = x^3 + 12x + 5$, in $[-3, 3]$.
9. When we say that a function $y = f(x)$ is concave up in $[a, b]$?
10. If f and g are two monic polynomials (leading coefficient is 1) of same degree, what is $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$?

11. What is Riemann sum for a function f on the interval $[a, b]$.
12. If $f(x) > 0$, what is the area of the region bounded by the the graph of f , the x -axis and the ordinates $x = a$ and $x = b$.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** questions (13-24).

Each question carries 2 mark.

13. Using formal definition of limit, show that $\lim_{x \rightarrow 1} (5x - 3) = 2$.
14. Using intermediate value theorem, show that there is a real number which is exactly one less than its cube.
15. Find left and right limits of the function f at $x = 2$, where $f(x) = \begin{cases} 3 - x & x \leq 2 \\ \frac{x}{2} + 1, & x > 2. \end{cases}$
16. Let $f(x) = -x^3 + 12x + 5$, $x \in [-3, 3]$. Where does the function f assume extreme values and what are these values ?
17. Define removable discontinuity and give an example.
18. Verify Rolle's theorem for the function $f(x) = (x - 2)(x - 3)$ on the interval $[2, 3]$.
19. Find the horizontal/vertical asymptotes of the graph of $f(x) = \frac{x^3 - 1}{x^2 - 1}$.
20. Find the average of $y = 2x - x^2$ in $[0, 3]$.
21. Find the linearization of $f(x) = 2 - \int_2^{x+1} \frac{9}{1+t} dt$.
22. Find dy/dx if $y = \int_x^1 \sqrt{1+t^2} dt$. Explain main steps in your calculation.

23. Find the area between $y = \sin x$, $x = -\pi/2$, $x = \pi/2$ and the x -axis.
24. Write down the main steps to find the volumes of solids by the method of slicing.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any six questions (25-33).

Each question carries 5 marks.

25. Define continuity and different types of discontinuity of a function $f(x)$ at a point a .
26. State Rolle's theorem and verify it for the function $f(x) = \frac{x^3}{3} - 3x + 2$ in the interval $[-3, 0]$.
27. State and prove L'Hospital's Rule (First form).
28. State Mean Value Theorem and verify for the function $y = 2x^3 - 3x^2$ in $[1, 2]$.
29. Evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \cot^2 x \right]$.
30. Express the solution of the following initial value problem as an integral. $y' = \tan x$, $y(1) = 5$.
31. If f is a continuous function on $[a, b]$, show that :
- $$(\min f) \cdot (b - a) \leq \int_a^b f(x) dx \leq (\max f) \cdot (b - a).$$
32. Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.
33. Find the volume, by slicing, of the solid which lies between planes perpendicular to the x -axis at $x = 0$ and $x = 4$. The cross sections perpendicular to the axis on the interval $[0, 4]$ are squares whose diagonals run from the parabola $y = -\sqrt{x}$ to the parabola $y = \sqrt{x}$.

(6 × 5 = 30 marks)

Turn over